Functional Requirements (FRs)

Blood Requests

FR1: The system shall allow a blood seeker to create a blood request.
FR2: The system shall send the blood request to the selected donor.
FR3: The system shall store and manage request information (request ID, seeker ID, donor ID, status, timestamp).
FR4: The system shall allow donors to accept or decline blood requests.
FR5: The system shall update the request status (pending, accepted, declined).

Notifications

FR6: The system shall notify donors when they receive a blood request.
FR7: The system shall notify blood seekers when a donor accepts or declines a request.
FR8: The system shall manage system notifications (e.g., admin alerts, updates).

Messaging

FR9: The system shall enable communication between donors and seekers once a request is accepted.
FR10: The system shall allow sending and receiving messages in real time.
FR11: The system shall store and display message history for both users.

Admin Dashboard

FR12: The system shall allow administrators to view and manage registered users.
FR13: The system shall allow administrators to manage hospitals and donation centers.
FR14: The system shall allow administrators to review reported accounts and take appropriate actions.
FR15: The system shall allow administrators to monitor and manage blood requests and platform activities.

Non-Functional Requirements (NFRs)

Performance

NFR1: Notifications shall be delivered within 5 seconds of a request being sent.
NFR2: The messaging system shall support at least 100 concurrent conversations without delay.

Security

NFR3: All messages shall be encrypted to protect user privacy.
NFR4: Only authenticated users shall be able to send or receive blood requests.
NFR5: Admin dashboard access shall be restricted to authorized administrators only.

Usability

NFR6: The request creation form shall be simple and intuitive, requiring no more than 3 steps.
NFR7: Notifications shall be displayed clearly in the user interface with distinct icons.
NFR8: The admin dashboard shall provide a clean interface with easy navigation menus.

Reliability & Scalability

NFR9: The system shall maintain 99% uptime during active hours.
NFR10: The system shall scale to support up to 10,000 registered users.